

Product Safety Report

AutoDrive X1 — Autonomous Driving Control Unit (ADCU)
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Executive summary

This report summarizes structural integrity, sensor suite performance, and regulatory alignment for the AutoDrive X1 ADCU under the latest NCAP-adjacent protocols and OEM validation cycles. Frontal offset and side-impact simulations indicate occupant compartment deformation within prescribed limits across the rated speed envelope, with cabin intrusion metrics meeting or exceeding baseline targets for the vehicle platform. LiDAR and camera fusion latency remained within specification during bench and track testing, including low-visibility scenarios with moderate precipitation.

Crash testing and structural assessment

Pendulum and barrier tests were conducted in accordance with internal safety matrices aligned to applicable regional requirements. Post-test inspection showed no anomalous fractures in primary load paths; supplemental restraint timing was verified against reference traces. The report package includes high-speed imaging and strain-gauge summaries used for correlation with finite-element models.

Sensors, software, and compliance posture

Environmental perception stacks were evaluated for false-positive and false-negative rates against curated datasets representing urban arterials, highways, and construction zones. Cybersecurity and functional-safety artifacts were reviewed for traceability to documented hazard analyses. Documentation for type-approval submissions is maintained under change control; this assessment does not replace statutory filings or OEM-specific certification obligations.